MATH329 · OPERATIONS RESEARCH

Midterm 461 تدريب

25 درجة · 10/10/2024

محارب

01 الأسئلة

حل كل سؤال في المساحة اللي تحته. الحل الكامل في ملف منفصل.

Q1 THEORY · 4 M

Answer the following:

a) The formal activities of Operations Research (OR) were initiated in _____ during World War II.

b) An optimization problem is called _____ when the objective function and all the constraints are linear functions of the decision variables.

c) To solve a LPP algebraically, we first put it in the Standard Form. What is this means?

GRAPHICAL · 6 M

Q2

Solve the following problem graphically:

Maximum $z = 10x_1 + 5x_2$

S.to: $x_1 - x_2 \leq 1$

$3x_1 + x_2 \geq 7$

$x_2 \leq 3$

$x_1 \geq 0, x_2 \geq 0$

STANDARD FORM · 6 M

Q3

Given the following problem:

Minimize $Z = 7x_1 + 3x_2$

S.to $2x_1 + x_2 \leq 8$

$x_1 - 3x_2 \geq 12$

$4x_2 = -3$

$x_1 \geq 0$, x_2 unrestricted in sign.

1. What methods can be applied to solve the given linear programming problem?

2. Write the problem in standard form (equation form). "Don't solve the problem"

STANDARD FORM + SIMPLEX · 9 M

Q4

Given the following problem:

Maximize $Z = 8x_1 + 7x_2$

S.to $20x_1 + 16x_2 \leq 62$

$-x_1 + x_2 \leq 1$

$x_1 \leq 3$

$x_1 \geq 0, x_2 \geq 0$

a) Write the problem in standard form (equation form).

b) Solve the problem by the simplex method (stop at the first iteration "Only two tables").

Basic	x_1	x_2	s_1	s_2	s_3	Sol.
Z						
Z						